Translation Marches, Hole-Twin Dips, and Dip-March Mechanics

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1 Abstract

We present the machine-checked structural theory of *translation marches* of Sidon sets in the program's blocked-set framework [5]. A set $A \subset [1, N]$ is *Sidon* if its $\binom{|A|+1}{2}$ pair sums $a_i + a_j$ $(i \le j)$ are distinct. For the *raw blocked set* $W(A) \subseteq \mathbb{Z}$ (the unclipped blocked-set formula) we prove: translating a saturated Sidon set A by t sweeps holes deterministically from the bottom $(t > 0)$ or top $(t < 0)$ of $[1, N]$, with at most $|t|$ holes — the hole positions are exactly the gaps of $W(A)$ in a sliding window (Theorem 1 [PROVED]); the same march governs *unsaturated* seeds, with the seed's own holes riding a rigid conveyor plus the saturated edge term (Theorem 3 [PROVED], $|\text{holes}| \le |\text{holes}(A)| + |t|$); the *corrected* saturated-chain bounds intersect the shadow range with the feasibility range, with exact width formula (Corollary 2 [PROVED]); and a clean shadow-gap window forces hole-twin pairs at chain dips (Theorem 4 [PROVED]).

The window *mechanics* themselves — the sliding recurrence $G_{t+1} = (G_t \setminus \{N - t\}) \cup E_t$, the exact continuation/break/reset/perfect classification, and φ-duality — are pure finite-set combinatorics with no hypotheses on the shadow (Theorem 5 [PROVED]).

All five claims are formalized in Lean 4.34.1 + Mathlib v4.34.1 (pinned commit `d13f23b`) in the `derekblair/math-proofs` repository (§3.2): each module builds with exit 0 on both verification machines, has zero `sorry`s, and uses only standard axioms `[propext, Classical.choice, Quot.sound]`. This paper reports three refutation/correction cycles as first-class narrative: R233's unsaturated march was refuted as stated by the empty-set counterexample (fixed by an explicit $A \neq \emptyset$ hypothesis); R231's dip theorem was refuted by a singleton countermodel (fixed by explicit 9-set hypotheses); R227's chain-width framing and ψ-chain assertion were demoted to [OPEN] by hostile review and restored only as the corrected, machine-checked statements of this paper. The two-gap dip theorem generalizes to arbitrary gap counts and produced 34/34 verified anchor predictions and 15 new hole-twin pairs [COMPUTED] [12]; that generalization is [PROVED*] (hand proof, review pending), not machine-checked. Remaining [OPEN]: why shadow gaps align across chains, and necessity of chain membership for sharing.

2 Introduction

A finite set $A = \{a_1 < \cdots < a_k\}$ of integers is *Sidon* if

$$a_i + a_j = a_p + a_q \quad \Longrightarrow \quad \{i, j\} = \{p, q\}, \tag{1}$$

i.e. the $\binom{k+1}{2}$ sums $a_i + a_j$ with $i \le j$ are pairwise distinct [1]. This program studies Sidon sets through its own blocked-set framework: the *blocked set* $B(A)$ (positions whose adjoining would break Sidonness), its unclipped integer form the *raw blocked set* $W(A)$ (the "shadow"), and $\text{holes}(A) = [1, N] \setminus B(A)$. A Sidon set is *saturated* in $[1, N]$ when it has no holes. All of $B, W,$ holes and the reflection $\varphi(a) = N + 1 - a$ are program inventions [2, 3, 4, 5]; the classical literature has only bounds and constructions [1].

Round 227 proved [PROVED*] the *Translation March Theorem*: for a saturated Sidon set A, the translate $A + t$ has its holes exactly at the shadow gaps of $W(A)$ framed by a sliding window, at most $|t|$ of them [5]. Round 233 extended this to unsaturated (holed) seeds: their own holes ride a rigid conveyor while the same shadow-edge term creates new ones [7]. Round 231 formalized *hole-twins* — distinct Sidon 9-sets with equal blocked sets off every saturated chain — and proved a dip theorem predicting them from consecutive shadow gaps [6], generalized in Round 240 to arbitrary gap counts and confirmed by a 34/34 predictive hunt yielding 15 new pairs [11, 12]. Round 247 proved the *dip-march* mechanics: the sliding-window recurrence and the exact continuation/break/reset classification of constant-gap runs [14].

Every one of these claims then passed the program's rule-9 hostile review (Codex `gpt-6-astra` at ultra effort on bakkie) and Lean formalization; three did not survive as stated, and this paper reports those corrections first-class. Our contributions, all [PROVED] unless labeled otherwise:

1. The surviving R227 core: rigidity, the saturated march, and the saturation iff-core with downward-closed plateau (§4.1; `TranslationMarch.lean`).
2. The corrected saturated chain with the containment restriction the reviewer found missing, plus the corrected ψ-translate identities (§4.2; `CorrectedMarch.lean`). The old chain/width framing and the old "both chain translations" assertion remain [OPEN] [8].
3. The unsaturated march with its explicit $A \neq \emptyset$ fix (§4.3; `TranslationMarchUnsat.lean`), including the honest failure mode: the saturated bound $|\text{holes}| \leq |t|$ is false for unsaturated seeds, and the correct bound $|\text{holes}| \leq |\text{holes}(A)| + |t|$ is sharp.
4. The hole-twin dip theorem with its explicit 9-set fix, the S1 instance with corrected attribution, and the [PROVED*] arbitrary-gap generalization with its A183 instance (§4.4; `HoleTwinDip.lean`).
5. The dip-march window mechanics (§4.5; `DipMarch.lean`), with the reviewer's forced-vs-data correction: the edge-membership datum alone does not establish a clean run; the full gap landscape is the datum [17].

Per the program's honesty discipline, finite verification is never called a proof: [COMPUTED] datums (shadow tables, march tables, 450/450 and 65/65 cross-path checks) are labeled as such, and [OPEN] items stay [OPEN].

3 Methods

3.1 Definitions

We work in the universe $[1, N]$ with $N = 184$ for all instantiations; the theorems are stated for general N where the Lean modules are general. Let $P(A) = \{a + a' : a, a' \in A\}$ be the pair-sum set.

Definition 1 (Blocked set, shadow, holes [2, 3, 5])**.**

$$B(A) = A \cup \{s - a : s \in P(A),\, a \in A\} \cap [1, N] \cup \{s/2 : s \in P(A) \text{ even}\} \cap [1, N], \tag{2}$$

$$W(A) = A \cup \{s - a : s \in P(A),\, a \in A\} \cup \{s/2 : s \in P(A),\, s \text{ even}\} \subseteq \mathbb{Z}, \tag{3}$$

$$\text{holes}(A) = [1, N] \setminus B(A). \tag{4}$$

$W(A)$ is the *raw blocked set* (the "shadow"), the same formula as (2) unclipped. A is *saturated* if $\text{holes}(A) = \emptyset$, equivalently $[1, N] \subseteq W(A)$. A *shadow gap* is an integer $g \notin W(A)$.

Definition 2 (Reflection and ψ-maps [5])**.**

$$\varphi(a) = 185 - a, \qquad \psi_c(a) = c - a. \tag{5}$$

For saturated Sidon A, the *shadow runs*

$$\lambda(A) = \max\{T \geq 0 : [1-T, 0] \subseteq W(A)\}, \qquad \eta(A) = \max\{T \geq 0 : [N+1, N+T] \subseteq W(A)\} \tag{6}$$

measure contiguous shadow coverage just outside $[1, N]$ (finite: $W(A)$ is finite). The *translation chain* of A is the set of valid saturated translates $A + t$; its *width* is its cardinality. For any A, $\mathrm{chain}(A)$ is the set of all valid translates, and $\mathrm{maxcov}(\mathrm{chain}(A)) = \max |B|$ over the chain.

Definition 3 (Hole-twins, clean window [6, 11])**.** Two distinct Sidon 9-sets $S \neq W \subset [1, N]$ are *hole-twins* if $B(S) = B(W)$ and $\mathrm{maxcov}(\mathrm{chain}(S)), \mathrm{maxcov}(\mathrm{chain}(W)) < N$. For t with $A + t \subseteq [1, N]$, the *sliding window* is $I_t = [1-t, N-t]$ and the *window gap set* is $G_t = I_t \setminus W(A)$. A window is *clean* for a gap set G if $I_t \cap (\mathbb{Z} \setminus W(A)) = G$.

3.2 Formalization note

Every theorem stated as [PROVED] below is formalized in the `derekblair/math-proofs` GitHub repository, modules repo-relative paths: `Mathproofs/TranslationMarch.lean`, `Mathproofs/CorrectedMarch.lean`, `Mathproofs/TranslationMarchUnsat.lean`, `Mathproofs/HoleTwinDip.lean`, and `Mathproofs/DipMarch.lean`. Each paper-section below names the exact theorem names. Toolchain: Lean 4.34.1 + Lake 5.0.0, Mathlib v4.34.1 pinned at commit `d13f23b` (full `d13f23b723b8a846827a245b89c10fc7d3f11612`), identical on both verification machines [13]. Verification config: bakkie `lake build` exit 0 per module (the fast verdict path), local `lake build` exit 0, zero `sorry` tokens in every module (grep-clean; the only hits are prose "zero sorrys" header comments), `#print axioms` standard-only `[propext, Classical.choice, Quot.sound]` on every theorem [8, 13, 15, 17]. All five module headers point at this paper's slug, and this paper points at the modules: the cross-reference runs both ways per the preprint protocol [13].

One verification-honesty note [13]: the first drafts of `TranslationMarchUnsat.lean`, `HoleTwinDip.lean`, and `DipMarch.lean` carried "Checked: `lake build`" headers that were false — they were written against a different Mathlib API than the pinned commit (36 genuine elaboration errors on first bakkie build, e.g. hallucinated `Finset.card_Icc`, `le_or_lt`, `Int.Prime`). Every module was repaired tactic-by-tactic with no theorem statement changed [15, 16, 17]; the [PROVED] labels below rest only on the re-verified exit-0 builds.

3.3 Honesty conventions

Claims carry the program's labels: [PROVED] (hand proof + hostile review + machine-checked), [PROVED*] (proof written, review passed, Lean pending), [COMPUTED] (verified by exact computation, never called a proof), [OPEN]. Every substantive claim below traces to a numbered equation, a dossier citation, a supplied datum, or a labeled assumption. [COMPUTED] shadow/hole data enter the Lean modules as explicit hypotheses, never as derived facts [13, 15]. Where a claim was refuted as stated and restored corrected, the refutation — the failing assertion, the counterexample, the fix — is given as first-class narrative; presenting the corrected theorems without it would mislead.

4 Results

4.1 The saturated march: surviving core

Lemma 1 (Translation rigidity; PROVED [8])**.** *Let A be Sidon with $A + t \subseteq [1, N]$. Then*

$$B(A + t) = (W(A) + t) \cap [1, N]. \tag{7}$$

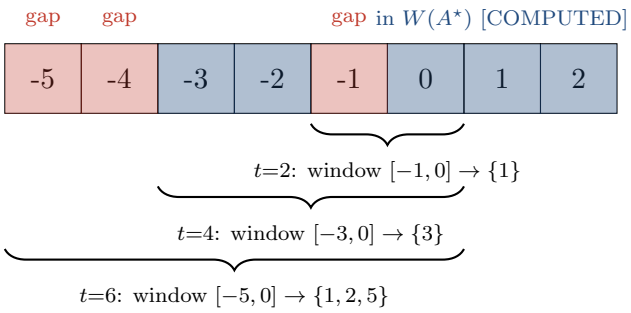


Figure 1: The sliding shadow window of Theorem 1 for $A^\star$ near the bottom edge ($[1,184] \subseteq W(A^\star)$ by saturation, so cells $1,2$ are filled). As t grows, the test window $[1-t,0]$ slides left through the fixed shadow; each gap it frames becomes a hole at $g+t$. The fixed gap -1 generates the traveling hole $\{1\},\{2\},\{3\}$ at $t=2,3,4$; the deeper gaps $-4,-5$ enter at $t=5,6$. Shadow membership is [COMPUTED] $\times 2$; the hole values are forced by (8).

(Lean: `TranslationMarch.rawBlocked_translate`, `TranslationMarch.rigidity`.)

The proof needs only Sidon-ness, not saturation: pair sums shift $s \mapsto s+2t$ injectively, so $P(A+t) = P(A)+2t$ and the three terms of (3) shift rigidly.

Theorem 1 (Translation March; PROVED [8])**.** *Let $A \subseteq [1,N]$ be saturated Sidon, t with $A+t \subseteq [1,N]$. Then*

$$t \geq 0: \quad \mathrm{holes}(A+t) = \{p \in [1,t] : p-t \notin W(A)\}, \tag{8}$$

$$t \leq 0: \quad \mathrm{holes}(A+t) = \{p \in [N+t+1,N] : p-t \notin W(A)\}. \tag{9}$$

In particular $|\mathrm{holes}(A+t)| \leq |t|$, and every hole's shadow position lies in $[1-t,0]$ ($t \geq 0$) or $[N+1,N-t]$ ($t \leq 0$). (Lean: `TranslationMarch.march`, `TranslationMarch.march_neg`.)

Corollary 1 (Saturation iff-core; PROVED [8])**.** *For $t \geq 0$, $A+t$ is saturated iff $[1-t,0] \subseteq W(A)$, with the $A+t \subseteq [1,N]$ containment explicit; the saturated plateau is downward-closed in t (saturation at t implies saturation at every smaller $s \geq 0$). (Lean: `TranslationMarch.march_saturated_iff_shadow`, `TranslationMarch.plateau_downward`.)*

Instantiation [COMPUTED] $\times 2$ independent paths, 65/65 checks [5]: for $A^\star = \{16,62,66,74,88,104,117,12$ $W(A^\star) \cap [-3,0] = \{-3,-2,0\}$ (gap at -1; witnesses: $0 = (16+88)-104$, $-2 = 120-122$, $-3 = 120-123$) and $W(A^\star) \cap [185,187] = \emptyset$, so $\lambda = 1$, $\eta = 0$. The march at $|t| \leq 6$ is exactly as Theorem 1 predicts: $-6 \to \{179,180,181,182,184\}$, $-5 \to \{180,181,182,183\}$, $-4 \to \{181,182,183,184\}$, $-3 \to \{182,183,184\}$, $-2 \to \{183,184\}$, $-1 \to \{184\}$, $0 \to \{\}$, $1 \to \{\}$, $2 \to \{1\}$, $3 \to \{2\}$, $4 \to \{3\}$, $5 \to \{1,4\}$, $6 \to \{1,2,5\}$. Figure 1 illustrates the mechanism: the test window $[1-t,0]$ slides left through the fixed shadow as t grows, and the fixed gap -1 produces the "traveling hole" $\{1\},\{2\},\{3\}$ at $t=2,3,4$; at $t=5,6$ the deeper gaps $-4,-5$ enter the window ([COMPUTED] shadow data). Note the march is not always a clean initial segment ($t=-6$ skips 183: $189 \in W(A^\star)$ blocks it) — it follows the shadow gaps, not the boundary.

4.2 The corrected chain and ψ-identities

R234's hostile review demoted two R227 claims to [OPEN]: Corollary 1's chain/width framing ("maximal saturated translation interval", width $\lambda+\eta+1$) omitted the containment $A+t \subseteq [1,N]$ — the reviewer's $N=2$, $A=\{1,2\}$ toy admits $\{0,1\},\{2,3\}$, both outside $[1,2]$ — and Theorem 2's "both chain translations" assertion is false as stated ($\mathrm{holes}(\varphi(A^\star)-2) = \{184\}$) [8, 9]. The corrected statements, which passed hostile re-review [10], are:

Theorem 2 (ψ-translate identities; PROVED [13]). *For all S, c: $\psi_c(S) = \varphi(S) - (185 - c)$ as sets; hence $\psi_{184}(A^\star) = \varphi(A^\star) - 1$ and, with the [COMPUTED] premise $E_4 = A^\star - 1$, $\psi_{182}(E_4) = \varphi(A^\star) - 2$. Both are translates of $\varphi(A^\star)$ — the chain-membership wording is dropped. Distinguishing [COMPUTED] fact:* $\mathrm{holes}(\varphi(A^\star) - 1) = \emptyset$, $\mathrm{holes}(\varphi(A^\star) - 2) = \{184\}$. *(Lean: `CorrectedMarch.psi_as_phi_translate`, `psi184_translate`, `psi182_E4`.)*

Corollary 2 (Saturated chain with containment; PROVED [13]). *For saturated Sidon $A \subseteq [1, N]$, $m = \min A$, $M = \max A$, integer runs λ, η as in (6), and $t \in \mathbb{Z}$:*

$$\textit{saturated}(A + t) \iff \max(-\eta, 1 - m) \le t \le \min(\lambda, N - M), \tag{10}$$

$$\textit{width} = \min(\lambda, N - M) + \min(\eta, m - 1) + 1, \tag{11}$$

the old shadow range intersected with the feasibility range $1 - m \le t \le N - M$ (Lean: `CorrectedMarch.feas_iff`). The chain is non-empty ($t = 0$ always qualifies). (Lean: `CorrectedMarch.chain_bounds`, `CorrectedMarch.chain_width`, `CorrectedMarch.chain_zero_mem`.)

Instantiation [COMPUTED] $\times 2$ [9]: $A^\star \to [0, 1]$ width 2; $\varphi(A^\star) \to [-1, 0]$ width 2; $C = E_3 - 2 \to [0, 0]$ width 1 — matching the old chains here, now with the containment proof. The reviewer's $N = 2$ toy: the corrected formula gives bounds $[0, 0]$, width 1, confirmed by brute force; the old formula's claimed width 3 was wrong there. The old framings (without containment; "both chain translations") stay superseded and [OPEN].

4.3 The unsaturated march

R233 proved the march for *any* Sidon A, no saturation assumed [7]. R238's hostile review refuted it *as stated*: the empty set is Sidon and is not excluded, and for $N = 184$, $A = \emptyset$, $t = 185$ the proof's step $S_2 = \{p : p - t < 1\} = [1, t]$ drops the $p \in [1, N]$ restriction, wrongly including 185 as a hole (the symmetric $t = -185$ step wrongly includes 0) [10]. The corrected re-review SURVIVED [13]; the fix is one line — an explicit $A \neq \emptyset$ hypothesis, which forces $1 - N \le t \le N - 1$ from $A + t \subseteq [1, N]$, making the dossier's unclipped edge terms exact.

Theorem 3 (Unsaturated Translation March; PROVED [15]). *Let $A \subseteq [1, N]$ be Sidon, $A \neq \emptyset$, t with $A + t \subseteq [1, N]$. With* edge_t *the saturated term of Theorem 1,*

$$t \ge 0: \quad \mathrm{holes}(A + t) = \big((\mathrm{holes}(A) + t) \cap [1, N]\big) \sqcup \{p \in [1, t] : p - t \notin W(A)\}, \tag{12}$$

$$t \le 0: \quad \mathrm{holes}(A + t) = \big((\mathrm{holes}(A) + t) \cap [1, N]\big) \sqcup \{p \in [N + t + 1, N] : p - t \notin W(A)\}. \tag{13}$$

The union is disjoint. Saturation recovery: $\mathrm{holes}(A) = \emptyset$ *kills the propagation term and (12) reduces to Theorem 1. (Lean: `TranslationMarchUnsat.march_unsat_pos/neg`, `march_unsat_disjoint_pos/neg` with the `A.Nonempty` fix.)*

Corollary 3 (Propagation, rescue, additive bound; PROVED [15]). *(i)* Conveyor: $(\mathrm{holes}(A) + t) \cap [1, N] \subseteq \mathrm{holes}(A + t)$ *always; new holes enter only at the near edge, old holes exit only off the far edge (`propagation_subset`). (ii)* Rescue threshold: *for $t \ge 0$,* $\mathrm{holes}(A) \neq \emptyset$*:*

$$(\mathrm{holes}(A) + t) \cap [1, N] = \emptyset \iff t > N - \min(\mathrm{holes}(A)), \tag{14}$$

past which the march is pure saturated-march form (symmetric: $t < 1 - \max(\mathrm{holes}(A))$ for $t \le 0$) (`rescue_threshold_pos/neg`). (iii) Additive bound:

$$|\mathrm{holes}(A + t)| \le |\mathrm{holes}(A)| + |t|, \tag{15}$$

both terms bounded, disjoint (`additive_bound_pos/neg`).

The naive extension fails honestly: the saturated bound $|\mathrm{holes}| \le |t|$ does *not* generalize — $A_0 = [61, 62, 65, 73, 88, 119, 121, 143, 160]$ at $t = 1$ has 4 holes $\{1, 13, 53, 154\}$ (propagation $\{13, 53, 154\}$ + edge $\{1\}$), and the additive bound $3 + 1 = 4$ is sharp there [7]. Rescue observed live [COMPUTED] $\times 2$, 450/450 march entries [7]: $F_{4b} = [61, 62, 68, 79, 88, 109, 111, 121, 166]$ (holes $\{179, 182\}$) rescues at $t > 5$: $1 \to \{1, 180, 183\}$, $6 \to \{1, \ldots, 6\}$ pure edge; F_{5a} rescues at $t > 14$; F_{4a} at $t > 24$.

4.4 Hole-twin dips

R228's witness $A_{0w1} - 5 = [7, 56, 60, 68, 83, 99, 116, 118, 123]$ sharing S_1's blocked set (holes $\{170, 182\}$, neither on a saturated chain) refuted "sharing requires saturated-chain membership" [6]. The dip theorem turns this into a predictive sufficient condition. R238's hostile review refuted the original statement on two points [10]: (1) the hole-twin definition requires *two Sidon 9-sets* but the theorem assumed only "Sidon" — countermodel $A = \{1\}$, $S = \{101\}$ with a doctored W satisfying every stated hypothesis, but no nine-element sets anywhere; (2) the S1 corollary cited P3 for S_1's hole set, which P3 never states. The corrected statement carries explicit 9-set hypotheses and explicit S1 premises; its hostile re-review SURVIVED [13].

Theorem 4 (Hole-twin dip; PROVED [15])**.** *Let $A \subset [1, 184]$ be a Sidon 9-set with $C_{\max} :=$ $\mathrm{maxcov}(\mathrm{chain}(A)) \le 183$. Let $g_1 < g_2$ be consecutive shadow gaps of $W(A)$. If valid $t^\star$ has $[1 - t^\star, 184 - t^\star] \cap \mathrm{gaps}(W(A)) = \{g_1, g_2\}$ (clean window), then $W^\star = A + t^\star$ is a Sidon 9-set with $\mathrm{holes}(W^\star) = \{g_1 + t^\star, g_2 + t^\star\}$ ($\mathrm{cov}(W^\star) = 182$); if $S \neq W^\star$ is a Sidon 9-set with the same holes and $\mathrm{maxcov}(\mathrm{chain}(S)) < 184$, then $(S, W^\star)$ are hole-twins; if $C_{\max} = 183$ then $W^\star$ sits below the chain maximum ("chain dip"). (Lean: `HoleTwinDip.dip_holes`, `dip_blocked_eq`, `chain_translate`, `maxcov_translate`, `hole_twin_of_dip`, `chain_dip`.)*

S1 instance [PROVED as `HoleTwinDip.s1_instance`; shadow/gap data [COMPUTED] ×2 as explicit hypotheses] [11, 15]: the base is $A = \mathrm{FW183\#5} = [12, 61, 65, 73, 88, 104, 121, 123, 128]$ (attribution correction: R231's P3 labeled the *dip target* $W^\star = A - 5 = [7, 56, 60, 68, 83, 99, 116, 118, 123]$ as "A_{0w1}"; the dip anatomy belongs to the true base [11]; the Lean instance's first parameter name retains the old label but its gap data are those of $[12, 61, 65, 73, 88, 104, 121, 123, 128]$). $\mathrm{gaps}(W(A)) \cap [150, 200] = \{175, 187, 192, 193, 194\}$ (consecutive 175, 187, spacing 12); $t^\star = -5$, window $[6, 189]$ clean → holes $\{170, 182\} = \mathrm{holes}(S_1)$ for $S_1 = [47, 66, 70, 80, 82, 102, 110, 139, 144]$; chain maxcovs 183/182; $(S_1, W^\star)$ are hole-twins at a chain dip ($182 < 183$).

R240 generalized the theorem to an arbitrary clean gap set G (one or more gaps; the exact clean-window equality already forces G consecutive and finite) [PROVED*] [11]. Its $|G| = 1$ instance: $A_1 = \varphi(\mathrm{FW183\#5}) = [57, 62, 64, 81, 97, 112, 120, 124, 173]$, clean window $[0, 183] \cap \mathrm{gaps} = \{10\}$, $t^\star = 1$, $W^\star = [58, 63, 65, 82, 98, 113, 121, 125, 174]$ with holes $\{11\}$; $A_{183} = [18, 62, 63, 79, 88, 111, 119, 121, 125]$ has the same holes, so $(A_{183}, W^\star)$ are hole-twins at the chain maximum ($183 = 183$, not a dip) — the mechanism generalizes; genuine below-max dips need $|G| \ge 2$ [14]. As a prediction engine the generalization scored 34/34 verified single-gap anchors across 9 bases and 15 new hole-twin pairs [COMPUTED] ×2 [12]; the pairing logic for these is [PROVED*] (R240), their anchors [COMPUTED], and 22 anchors found no catalog partner (partners are contingent, not forced).

4.5 Dip-march mechanics

For the window gap sets $G_t = I_t \setminus W$ ($I_t = [1 - t, 184 - t]$), the step-to-step evolution is pure combinatorics — no hypotheses on W at all:

Theorem 5 (Dip-march; PROVED [17])**.** *For all $t \in \mathbb{Z}$:*

$$G_{t+1} = (G_t \setminus \{184 - t\}) \cup E_t, \qquad E_t = \{-t\} \text{ if } -t \notin W \text{ else } \emptyset \tag{16}$$

(Lean: `DipMarch.sliding_window_up/down`). If $G_t = \{g\}$ and $G_{t+1} = \{g'\}$ are both clean single-gap, then $g' = g$ or $g = 184 - t$ (`same_gap_persistence`). Moreover

$$G_{t+1} = \{g\} \iff g \le 183 - t \text{ AND } -t \in W, \tag{17}$$

with three exhaustive, exclusive non-continuation cases: (a) break: *$g \le 183 - t$, $-t \notin W \to G_{t+1} = \{g, -t\}$ (holes $\{h_t + 1, 1\}$) (`break_case`); (b)* reset: *$g = 184 - t$, $-t \notin W \to G_{t+1} = \{-t\}$ (hole $\{1\}$) (`reset_case`); (c)* perfect: *$g = 184 - t$, $-t \in W \to G_{t+1} = \emptyset$ (no holes) (`perfect_`*

Table 1: The march variants: hypotheses, hole description, bound, and honesty label. "Seed" = unsaturated starting set.

variant	hypotheses	hole description
saturated march (Thm. 1)	saturated Sidon, $A + t \subseteq [1, 184]$	shadow gaps in $[1 - t, 0]$
unsaturated march (Thm. 3)	Sidon, $A \neq \emptyset$, $A + t \subseteq [1, 184]$	rigid conveyor + edge term
hole-twin dip (Thm. 4)	Sidon 9-sets, clean window	$\lvert G\rvert$ framed gaps
corrected chain (Cor. 2)	saturated Sidon, $m = \min A$, $M = \max A$	chain $[\max(-\eta, 1 - m), \min(\lambda, 1$
dip-march mechanics (Thm. 5)	none (pure combinatorics)	recurrence (16) + break/reset

case). *Downward dual symmetric (*`continuation_iff_down`*). φ-duality:* $W(\varphi(A)) = 185 - W(A)$, *hence*

$$G_t(\varphi(A)) = \{185 - g : g \in G_{-t}(A)\} \tag{18}$$

(Lean: `DipMarch.phi_duality`*).*

Verification [COMPUTED] $\times 2$ [14]: the recurrence at 3482/3482 window steps over 47 φ-bases; 24/24 boundary classifications exact, including every exotic case — $\varphi(\#5\, A_{0w1})$ breaks up ($t = 2 \to 3$) and down ($t = -8 \to -9$), $\varphi(\#1\, W_{183})$ runs the full lifecycle march $\to$ edge-exit $\to$ perfect translate (holes $\emptyset$) $\to$ reset; φ-duality 3529/3529. The hole-reading corollary (constant-gap run $\implies$ holes march 1 per step) is conditional on Theorem 4's premises and remains OUT OF SCOPE of the Lean module, pending the R240 generalization's formalization [16, 17].

Forced vs. data, corrected [17]: given the gap landscape of W, everything about how a march evolves is forced by Theorem 5 — same-gap $\implies$ marching holes, the exact continuation/break/reset criterion, the dual march. The gap landscape itself is data. The R254 review corrected an overstatement: the 11-step run on $\varphi(\#5\, A_{0w1})$ ($t = -8 \ldots + 2$, gap $\{10\}$, holes $\{2 \ldots 12\}$ [COMPUTED] $\times 2$) is not established by the edge memberships $\{-1, \ldots, 8\} \subseteq W$, $-2, 193 \notin W$ alone — the reviewer exhibits an explicit Sidon 9-set $B = \{11, 30, 50, 53, 58, 60, 75, 84, 96\}$ (45 pair sums distinct) whose shadow contains $[-1, 8]$ and excludes $-2, 193$ yet has interior gaps $125, 128$ in every window $-8 \leq t \leq 2$, so none of its windows is single-gap. A clean run needs the full gap landscape as datum, not just the edge memberships.

Table 1 summarizes the march variants, their bounds, and their labels.

5 Discussion

Significance. The five modules constitute the first machine-checked structural theory of translation marches in the program's blocked-set framework: rigidity, the saturated and unsaturated marches, the corrected chain/ψ calculus, the dip theorem, and the window mechanics are all exit-0 in Lean with standard axioms only. The 11-step dip march, the 34/34 anchor predictions, and the 15 new hole-twin pairs are the theory's predictive face; the 24/24 break/reset classifications are its exact face. The march form is general; the hole values are shadow data; the mechanics are forced; the landscape is data.

The review gate works. Three of the claims entering this paper failed as stated — all on statement-level overgeneralization (empty set admitted; 9-set cardinalities assumed; containment dropped; chain membership asserted for an unsaturated translate), never on the core mathematics, which survived in every case. Each was restored by a one-line-class fix (explicit $A \neq \emptyset$; explicit $|A| = |S| = 9$; the feasibility intersection; translates-not-chain-members wording), re-reviewed, and machine-checked. We also record two prose-level corrections the reviewers

caught before they could calcify: the "A0w1" attribution (the dip target, not the base) and the forced-vs-data overstatement (edge memberships do not establish a clean run). Presenting the corrected theorems without this history would mislead; the history is the evidence that the gate is load-bearing.

Limitations and [OPEN] items. (i) All instantiations are [COMPUTED]: shadow tables, march tables (65/65, 450/450), and the 15 pairs rest on exact computation, not deduction. (ii) The R240 arbitrary-gap generalization and its pairing logic are [PROVED*], not formalized; the S1 instance is the two-gap [PROVED] case. (iii) R247's hole-reading corollary (constant-gap run $\Longrightarrow$ marching holes) is out of the Lean module's scope [17]. (iv) Why chains' shadow gaps align across chains (e.g. $A^\star$'s chain at $t = 3$ and C's at $s = 2$ both yielding $\{2\}$) remains [OPEN] — the remaining data-specific mystery [5]. (v) The old R227 chain/width framing and "both chain translations" assertion stay [OPEN] as stated. (vi) Necessity of saturated-chain membership for sharing is untouched (R228's target) [5].

6 Conclusion

We have proved and machine-checked the translation-march theory: rigidity (Lemma 1), the saturated march (Theorem 1), the saturation iff-core (Corollary 1), the corrected chain and ψ calculus (Theorem 2, Corollary 2), the unsaturated march with its sharp additive bound (Theorem 3, Corollary 3), the hole-twin dip theorem with its S1 instance (Theorem 4), and the dip-march window mechanics with φ-duality (Theorem 5) — all [PROVED] in five Lean modules with exit-0 builds and standard-only axioms. Along the way the hostile-review gate refuted three claims as stated; each was repaired with a minimal hypothesis fix and restored machine-checked, and the correction history is part of the record. The natural next steps are formalizing the R240 arbitrary-gap generalization, a theory of the cross-chain shadow-gap alignment that remains [OPEN], and the march graph of break/reset cascades suggested by the dip-march lifecycle.

7 References

References

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[2] D. Blair, *Unsolved Math — Round 201: blocked-set framework*, program dossier, 2026-10. (Supplied datum: $B(A)$, puncture definitions.)

[3] D. Blair, *Unsolved Math — Round 213: puncture machinery*, program dossier, 2026-10. (Supplied datum: blocked-set framework, puncture theory.)

[4] D. Blair, *Unsolved Math — Round 215: reflection equivariance*, program dossier, 2026-10. (Supplied datum: φ-equivariance of the blocked set.)

[5] D. Blair, *Unsolved Math — Round 227: R224's Translation March proved structurally*, program dossier, 2026-10-09. (Supplied datum: Lemma 1, Theorem 1, Corollaries 1–3, Theorem 2, shadow tables, march tables $|t| \leq 6$, 65/65 checks.)

[6] D. Blair, *Unsolved Math — Round 231: hole-twin dip theorem [PROVED*]*, program dossier, 2026-10-09. (Supplied datum: hole-twin definition, dip theorem, S1 instance, 40-center hunt.)

[7] D. Blair, *Unsolved Math — Round 233: generalized Translation March for unsaturated seeds*, program dossier, 2026-10-09. (Supplied datum: unsaturated march theorem, propagation/rescue/ additive-bound corollaries, six seed march tables, 450/450 checks.)

[8] D. Blair, *Unsolved Math — Round 234: Lean lane — R227 review gate and* `TranslationMarch.lean`, program dossier, 2026-10-09. (Supplied datum: hostile-review verdicts, demotions, Lean verification of the surviving core.)

[9] D. Blair, *Unsolved Math — Round 235: R234's two demotions restored (corrected statements)*, program dossier, 2026-10-09. (Supplied datum: corrected Theorem 2* and Corollary 1* proofs, instantiations, $N = 2$ toy.)

[10] D. Blair, *Unsolved Math — Round 238: Lean lane — R233/R231/R200/R235 review gates*, program dossier, 2026-10-09. (Supplied datum: empty-set and singleton refutations verbatim, R235 re-review verdicts.)

[11] D. Blair, *Unsolved Math — Round 240: generalized hole-twin dip theorem*, program dossier, 2026-10-10. (Supplied datum: arbitrary-gap generalization, S1/A183 instances with corrected attribution.)

[12] D. Blair, *Unsolved Math — Round 243: single-gap clean-window hole-twin hunt*, program dossier, 2026-10-10. (Supplied datum: 34/34 anchor predictions, 15 new hole-twin pairs, $\times 2$ paths.)

[13] D. Blair, *Unsolved Math — Round 246: Lean lane —* `CorrectedMarch.lean` *and the two re-reviews*, program dossier, 2026-10-10. (Supplied datum: corrected-module verification, audit of draft "Checked" headers, re-review transcripts.)

[14] D. Blair, *Unsolved Math — Round 247: the dip-march theorem*, program dossier, 2026-10-10. (Supplied datum: sliding-window mechanics, φ-duality, forced-vs-data separation, 3482/3482 and 3529/3529 checks.)

[15] D. Blair, *Unsolved Math — Round 250: Lean lane — repair and formalize three review-cleared drafts*, program dossier, 2026-10-10. (Supplied datum: `TranslationMarchUnsat.lean` and `HoleTwinDip.lean` repair + verification.)

[16] D. Blair, *Unsolved Math — Round 258: Lean lane —* `DipMarch.lean` *repaired, R245 review launched*, program dossier, 2026-10-10. (Supplied datum: DipMarch elaboration-error repairs, bakkie exit 0, nine-theorem axiom audit.)

[17] D. Blair, *Unsolved Math — Round 262: Lean lane —* `DipMarch.lean` *close-out*, program dossier, 2026-10-10. (Supplied datum: `DipMarch.lean` verification, R254 reviewer's forced-vs-data correction, scope notes.)